Month 3 –Documentation

Name: Namrata Dolas

(Hardware Design Intern)

Reporting Manger / Senior: Balaji Mayavan

Date: 2 Feb 2026

1. Wire Harness Designing (Wire Viz)

- Completed Distribution Panel Wire Harness Designing using WireViz.
- Worked on Operating Panel Wire Harness Designing using WireViz (v0.4.1).
- Installed and set up WireViz and the required Python environment.
- Completed a mini task to understand WireViz YAML syntax.
- Created and completed the YAML file for the Distribution Panel.
- Generated and updated the Bill of Materials (BOM) for GE wiring using WireViz.
- Gained an overview of the WireViz setup and workflow for harness designing.

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2. Schematic Design & Documentation (Cadence OrCAD Capture)

- Installed and set up Cadence OrCAD Capture software.
- Studied and referred AerX_GE Wiring Schematics.
- Updated GE Wiring Schematics, including:
    - Separation of AC and DC supplies for LED power.
    - ERC (Electrical Rule Check) verification.
- Completed GE Wiring schematic in Cadence Capture.
- Generated BOM (Bill of Materials) from updated schematics.
- Worked on GE Wiring Diagram documentation.

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3. PCB Design Exposure (Cadence Allegro)

- Took an overview of PCB Designing in Cadence Allegro.
- Explored and documented important Allegro command panel commands for smooth PCB design workflow.
- Installed and verified Spice models as per updated DVRU BOM (approximately 29–30 models).
- Updated and maintained the Spice Models Checklist.

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4. Product Research & Ideation

- Completed Phase-I (Research & Ideation) of the Product Life Cycle for the Test PCB, referring to the Aerx Test Jig PCB. • Studied and analysed:
    - DI (Digital Input) Section
    - DO (Digital Output) Section
    - AI (Analog Input) Section
    - Ethernet Protocol
- Explored basic computer networking concepts relevant to Ethernet, including:
    - Star topology
    - Mesh topology
    - Other basic network structures

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5. Power Analysis & Industrial Documentation

- Studied Power Analysis fundamentals, including:
    - DC systems
    - AC systems
    - Single-phase and Three-phase

      power
- Explored industrial-standard documentation formats using:
    - MDPI
    - Google Scholar
    - IEEE
- Referenced these resources for AerX_GEN5_Cart_Power_Analysis Report preparation.

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6. Electronics Fundamentals & Circuit Design

- Worked on Audio Panel Schematic, updating:
  - DO section using PC817 optocoupler and BC547 BJT.
- Revised Analog Electronics topics:
  - Diode fundamentals
  - BJT fundamentals
- Studied:
  - Semiconductor physics
  - Construction and working of diodes
  - Diode breakdown mechanisms
  - Junction capacitance
  - Diode modelling
  - AC analysis and testing of diodes

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7. Simulation & System Exposure

- Explored Flight Simulation concepts.
- Studied the AerX 'D' Flight Simulator.
- Explored HTT40 RVCD, using reference documents shared by Senior Balaji.

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8. Reviews & Team Coordination

- Completed feedback and review process with:
- Manager Jagdeesh
- Reporting Manager / Senior Balaji
- Discussed progress, learning outcomes, and improvements from the last two months.
- Completed Month-2 Documentation as part of reporting requirements.